Standard Designs Documentation

There are three available standard designs:
1. Randomize-All Design: All eligible participants are randomized to treatment vs. control regardless their biomarker status. The trial tests whether the treatment is efficacious across all participants.
2. Enrichment Design: All eligible participants are first screened for the selected biomarker. Only participants who screen positive for the biomarker (M+) are randomized to treatment vs. control. The trial tests whether the treatment is efficacious in M+ participants only.
3. Strategy Design: All eligible participants are first screened for the selected biomarker. Participants are randomized to either a “biomarker-based strategy” (where M+ and M- participants receive the experimental or control treatments, respectively) or to a “biomarker-agnostic strategy” (where all participants receive the control treatment). The trial tests whether using the biomarker-based strategy to guide treatment improves outcomes.

Standard Designs

Inputs Panel

Sample size of arm1 := The number of participants in arm 1 (control).
Targeted power := designated $1-\beta$ level for the entire trial. One minus the type II error rate.
Standardized effect size for M+ patients (Δ_+) := The expected standardized effect size for biomarker positive population.
Standardized effect size for M- patients (Δ_-) := The expected standardized effect size for biomarker negative population.
Biomarker prevalence := proportion of general population that are biomarker positive.
Hypothesis Test := Two-sided tests evaluate a null hypothesis of equality, while one-sided tests evaluate a null hypothesis with a greater-than or less-than condition.
Significance level (α) := designated α level for the entire trial, false positive rate overall.

Randomization ratio (Arm 2 / Arm 1) := Allocation ratio of arm 2 (strategy/treatment) to arm 1 (control).
Accrual rate of M+ participants (pts/month) := Accrual rate of biomarker-positive (M^+) individuals per month.
Accrual rate of M- participants (pts/month) := Accrual rate of biomarker-negative (M^-) individuals per month.

Inputs:

Select your design

- (●) Randomize-All
- ( ) Enrichment
- ( ) Strategy

What would you like to compute?

- ( ) Power
- (●) Sample Size

Targeted power

0.8

Standardized effect size for M+ patients (Δ_+)

0.5

Standardized effect size for M- patients (Δ_-)

0.1

Biomarker prevalence

0.33

Test type

- (●) One-sided
- ( ) Two-sided

Significance level (α)

0.05

Randomization ratio (arm2/arm1)

1

M+ Accrual rate (individuals/month)

5

M- Accrual rate (individuals/month)

15

Calculate Now!

Click **Calculate Now!** to view outputs.

Outputs Panel

Current Design

An illustration produced to show the current design of the trail based on your inputs. To copy, simply right click>"copy image"or"save image as". If texts are not fitting well, please adjust the width of you browser until it fits properly.

Design Summary

A test summary of current design of the trial based on your inputs. You can directy copy it since it is in text format.

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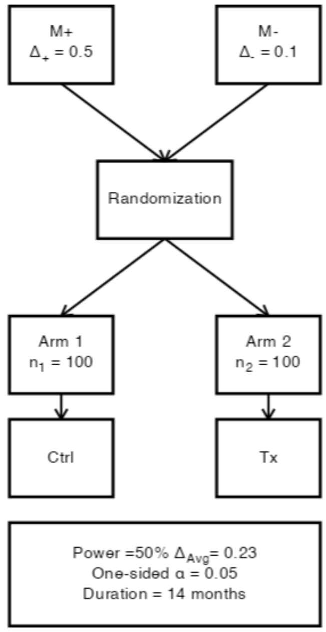
Current Design



Design Summary: Estimating power for proposed trial with selected sample sizes

This biomarker-informed trial follows a randomize-all design, in which participants are enrolled without regard to biomarker status. Both biomarker-positive (M+) and biomarker-negative (M-) individuals are randomized in equal proportions into two arms: Arm 1 (control) receives the control intervention, and Arm 2 (treatment) receives the experimental therapy. No stratification or restriction is applied based on biomarker status.

The null hypothesis (H_0) posits that there is no difference in the primary outcome between the treatment and control arms in the overall study population, irrespective of biomarker status. Rejecting H_0 would suggest that the experimental therapy is effective in the general population, providing evidence that biomarker status is not required to demonstrate treatment benefit. The standardized treatment effect for M+ and M- individuals is $\Delta_+ = 0.5$, and $\Delta_- = 0.1$, respectively. The estimated average effect size difference between trial arms is $\Delta_{Avg} = 0.232$.

Power and Total Sample Size

A minimum of 100 participants in the treatment arm and 100 participants in the control arm achieves 50% statistical power (one-sided type 1 error rate $\alpha = 0.05$). Based on a biomarker prevalence of 0.33, the expected sample size is projected to be 66 for M+ individuals and 134 for M– individuals.

Trial Duration

Assuming monthly recruitment rates of 5 M+ and 15 M– participants, the estimated trial duration is 14 months.

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